Individual Differences in Implicit Learning Measured by Reaction Time Variability

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1. Introduction

Implicit Learning[1]

People acquire statistical regularities without concious awareness

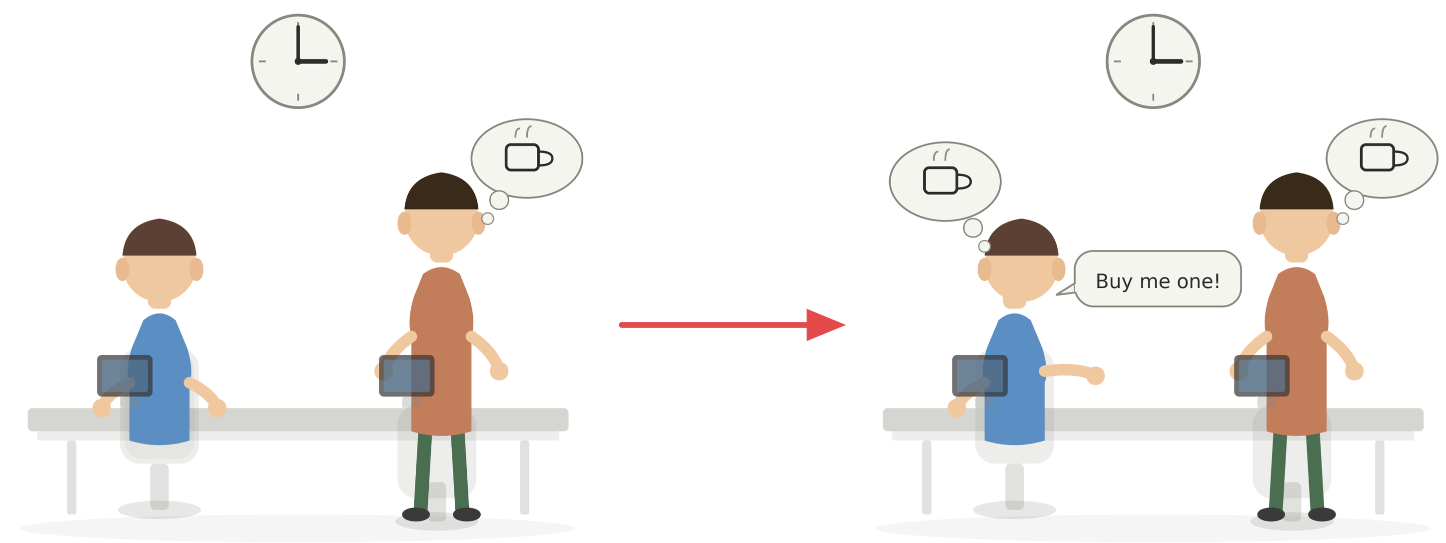
- Implicit learning shows **marked individual differences**[2], but the underlying factors remain poorly understood.

Objective

Examine whether attentional variability regulates the progression of implicit learning

Approach

Index attentional state via reaction time variability (CV of log-RT)

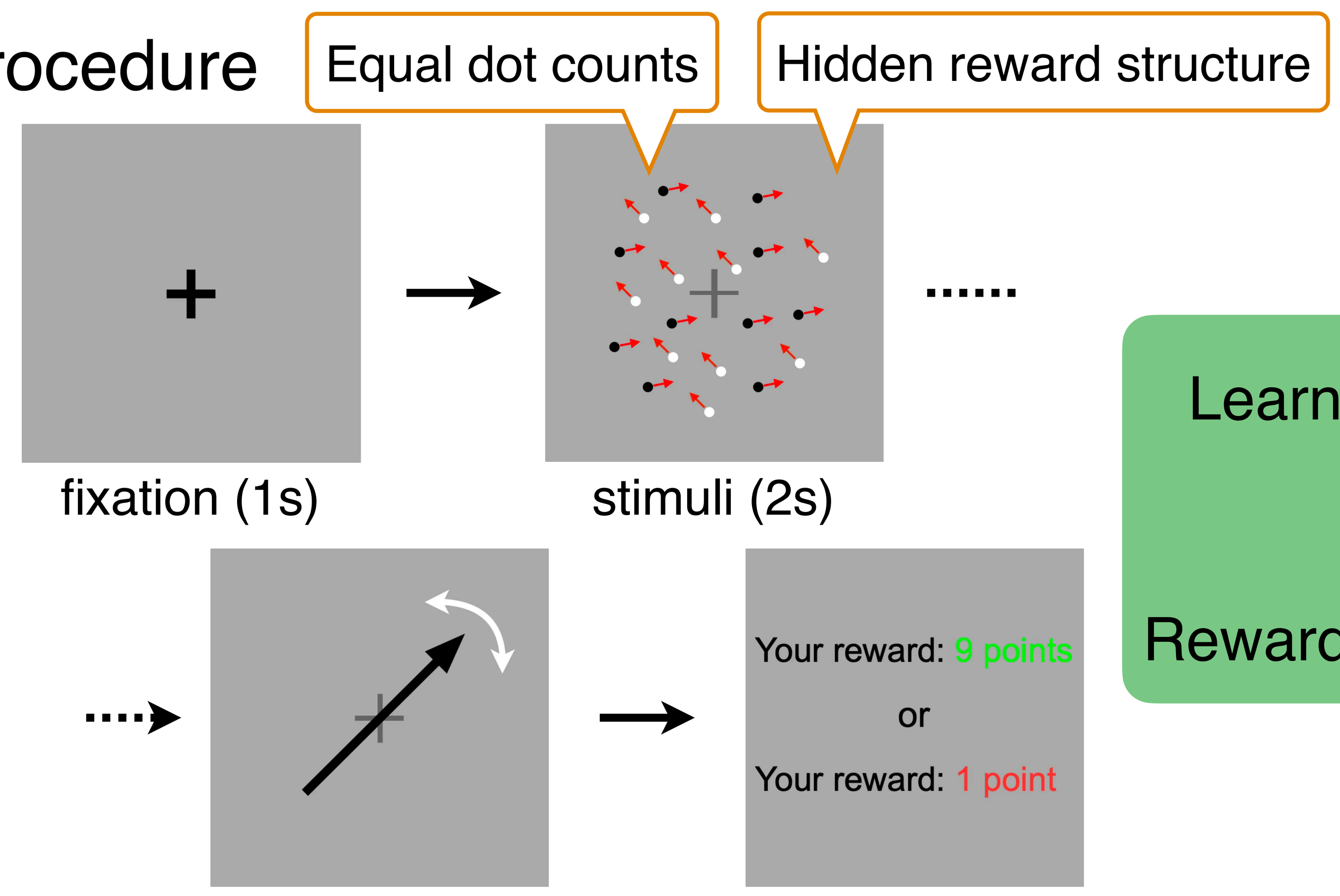


In IoT-rich environments, human attention — a finite and variable resource — is constantly demanded.

Understanding implicit learning is key to designing systems that align with human cognitive characteristics.

2. Experiment

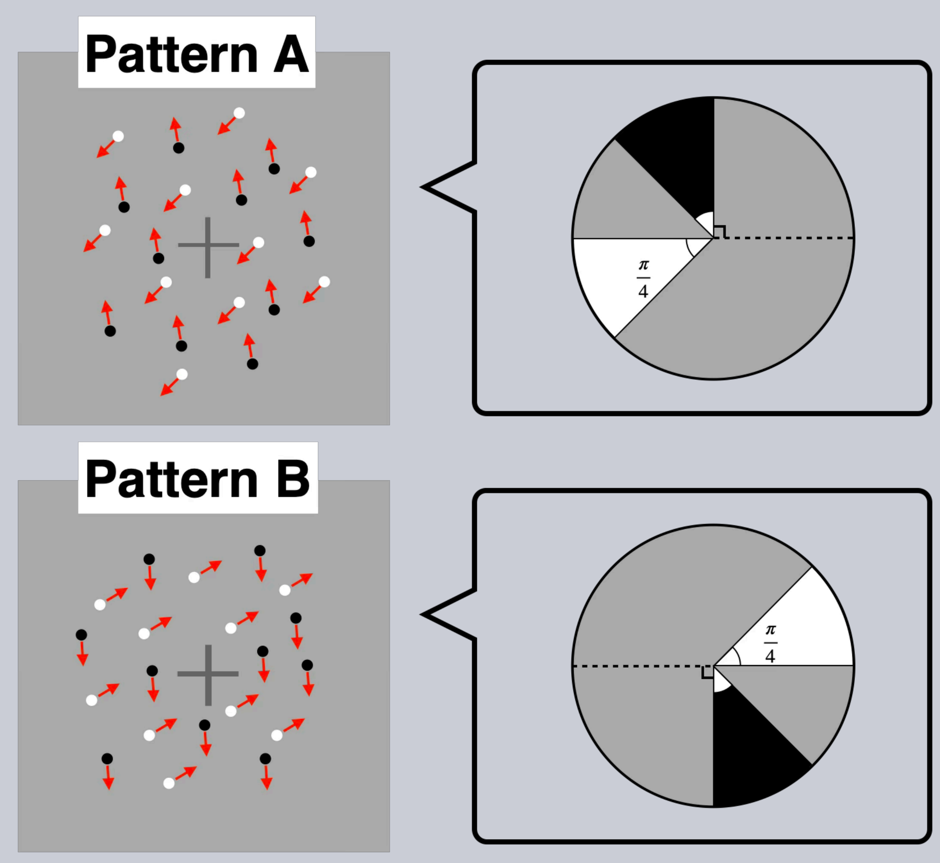
Procedure



Learning structure

Reward maximization

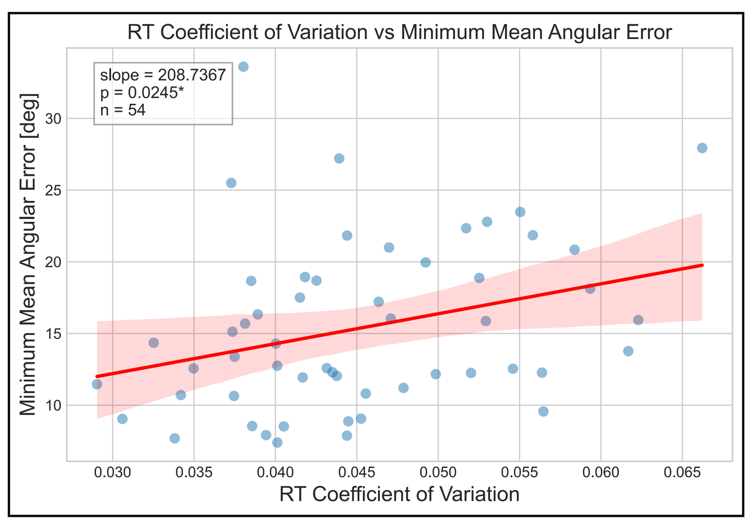
Reward structure



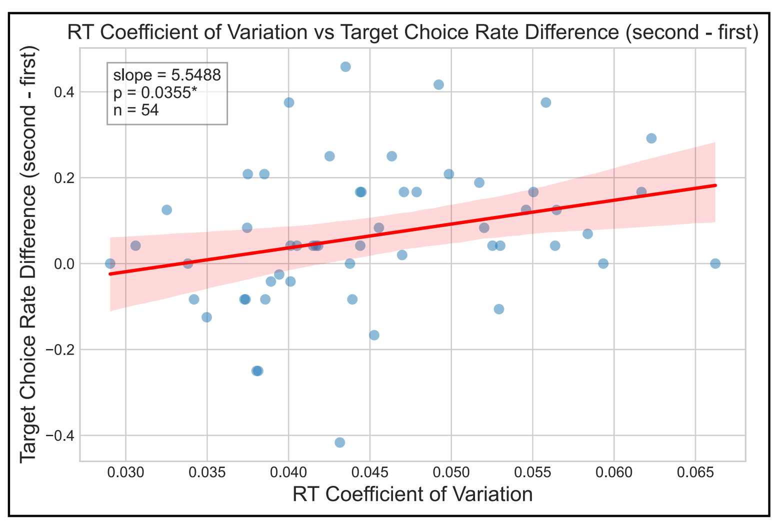
Pattern A — **White dots：low rewards**, **Black dots：high rewards**

Pattern B — **White dots：high rewards**, **Black dots：low rewards**

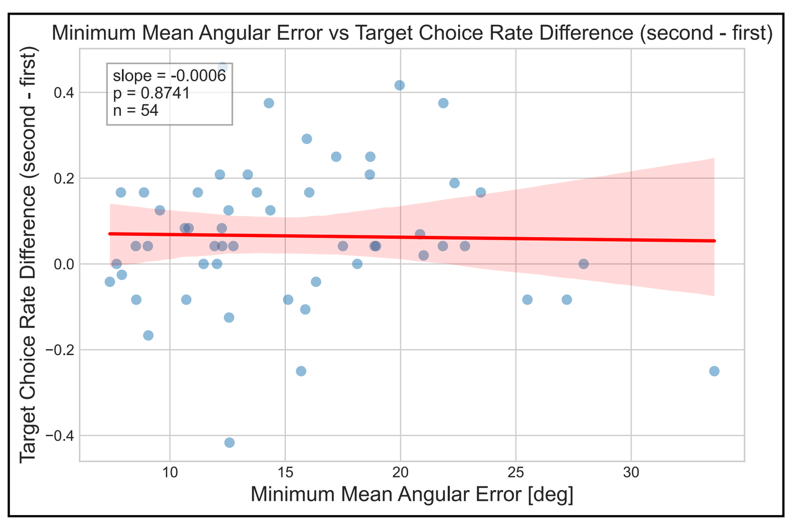
3. Result



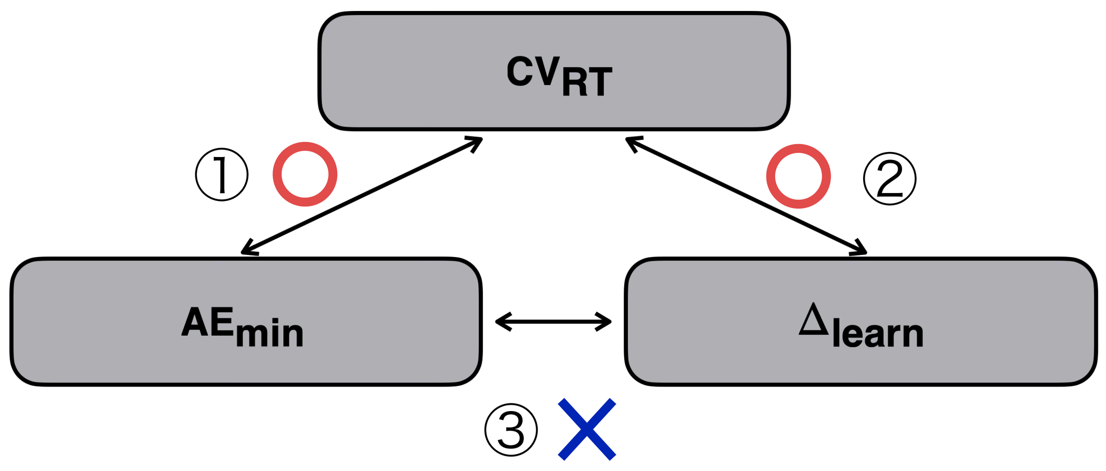
① $CV_{RT} \longleftrightarrow AE_{min}$



② $CV_{RT} \longleftrightarrow \Delta_{learn}$



③ $AE_{min} \longleftrightarrow \Delta_{learn}$



CV_{RT} (Coefficient of variation of RT) → Attentional variability

AE_{min} (Minimum Angular Error) → Perceptual precision

Δ_{learn} (Target choice rate difference) → Learning progression

Greater attentional variability predicted learning progression, independent of perceptual precision.

4. Conclusion

- RT variability predicts lower perceptual precision yet greater learning progression.
- The RT variability–angular error relationship supports RT variability as an index of attentional instability.
- RT variability reflects the initial conditions of learning rather than learning ability itself.
- Attentional lapses may facilitate implicit processing, consistent with mind-wandering research[3].
- Results partially support the hypothesis; trial-level attentional indices are needed in future work.

Reference

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[2] P.B. Kalra, J.D. Gabrieli, and A.S. Finn, “Evidence of stable individdual differences in implicit learning,” Cognition, vol.190, pp.199–211, 2019.
[3] A. Decker, M. Dubois, K. Duncan, and A.S. Finn, “Pay attention and you might miss it: Greater learning during attentional lapses,” Psychonomic Bulletin & Review,